Smears–Vohralík equilibration extended to convection: what was implemented

galerkit, gap 7 of afc_aposteriori_estimators-1.md. Code: `src/galerkit/convection_diffusion/sv_estimator.py`; tests: `tests/test_cdr_sv_estimator.py`.

1. In one paragraph

[SV20] (Smears & Vohralík, M2AN 54, 2020) give a guaranteed, robust, locally efficient equilibrated-flux estimator for $-\varepsilon^2\Delta u + \kappa^2 u = f$. galerkit now has the analogue for $-\varepsilon\Delta u + b\cdot\nabla u + cu = f$, discretised by any conforming P1 method. It works in 2D and 3D, for every algebraic stabilisation (AFC, MUAS, SMUAS, BBK, ...), and at inexact nonlinear iterates.

- **The upper bound** is guaranteed: it needs only $\sigma > 0$ and no Galerkin orthogonality. It bounds the energy norm, the augmented norm $|||e|||_\oplus = |||e||| + \|B_A(e,\cdot)\|_*$, and the gap 2 measure $E = |||e|||_\oplus + \|S^\sim\|_*$.
- **Numerically**, it is robust in ε for the augmented norm: effectivity 3.2–7.3 for $\varepsilon = 10^{-2} \ldots 10^{-8}$, against up to 280 for the classic ESV bound.
- **Efficiency** (global, robust, in E) has a proof sketch built from SV20 Lemma 4.2 and Verfürth 2005 Thm. 4.1. It is not refereed.

2. What is in the code

item	where
`sv_estimator(S, u, op, b, sigma, flux=None, D_mode=("direct", "A"), w_const=inf)`	`sv_estimator.py`
RTN_1 basis on simplices ($P_1^d + y\cdot P_1$, $y = (x - x_K)/h_K$), `rt1_basis`	`sv_estimator.py`
All vertex-patch problems in one block-diagonal sparse KKT system (broken RTN_1 + normal-moment constraints), `_solve_patches`	`sv_estimator.py`
`estimator="sv"` in `AdaptiveAFC` (marking and comparison); rows `eta_sv`, `eta_sv_aug`, `eta_sv_E`, `eff_aug_sv`, `eff_E_sv`	`adaptive_afc.py`
`--estimator sv` in `aj27` and `adaptive3d`; `sv` among the compared estimators of `guaranteed3d` ($\sigma > 0$ problems)	CLIs
`A_tensor` (cellwise artificial-diffusion tensor) returned by `abr17_estimator`	`flux_estimator.py`

3. The construction

Notation. The energy norm is $|||v|||^2 = \varepsilon\|\nabla v\|^2 + \sigma\|v\|^2$, with $0 < \sigma \le c - \tfrac{1}{2}\mathrm{div}\, b$. The residual is $r = f - b\cdot\nabla u_h - cu_h$. The diffusion tensor D is εI ("direct") or $\varepsilon I + A$ ("A-version": A is the stabilisation tensor of gap 2, so the functional is $\mathrm{Res}_A = \mathrm{Res} - S^\sim$).

Step 1 (= SV20 Def. 2.3 with the data $f^\sim = f - b\cdot\nabla u_h - (c - \sigma)u_h$ and $\kappa^2 = \sigma$). After the potential is eliminated through the constraint, every vertex a solves a weighted H(div) least-squares problem:

$\sigma^a = \mathrm{argmin}_{v \in V^a} w_a^2\, \varepsilon^{-1}\|\psi_a D\nabla u_h + v\|^2_{\omega_a} + \sigma^{-1}\|\mathrm{div}\, v - g_a\|^2_{\omega_a}$, $g_a = \Pi_1(\psi_a r - \nabla\psi_a\cdot D\nabla u_h)$.

- **Space.** V^a is RTN_1 on the patch.
  - The normal component is 0 on the patch boundary, except on Dirichlet faces of a boundary vertex, where it is free.
  - On Neumann faces it equals $-\Pi_F(\psi_a g_N)$.
- **Weight.** The default is $w_a = 1$; `w_const` gives SV's $w_a = \min\{1, C(\sqrt{\varepsilon/\sigma}/h_a)^{1/2}\}$.
- **Assembly.** $\sigma_h = \Sigma_a \sigma^a$ lies in H(div), and $\Sigma_a g_a = \Pi_1 r$.

Step 2 (estimator). With $\tau = D\nabla u_h + \sigma_h$ and $m_K = \min\{h_K/(\pi\sqrt{\varepsilon}), \sigma^{-1/2}\}$:

$\eta_K = \min\{\varepsilon^{-1/2}\|\tau\|_K, \Sigma_F c_F\|\tau\cdot n\|_F + m_K\|\mathrm{div}\,\tau\|_K\} + \sigma^{-1/2}\|\mathrm{div}\,\sigma_h - \Pi_1 r\|_K + m_K\|r - \Pi_1 r\|_K$,

$c_F^2 = |F|/|K| \cdot \min\{h_K^2(1/\pi^2 + 2/(d\pi))/\varepsilon,\ 1/\sigma + h_K/(d\sqrt{(\sigma\varepsilon)})\}$.

Step 3 (bounds).

- $\eta_{dir} = (\Sigma_K \eta_K^2)^{1/2}$ with $D = \varepsilon I$, and η_A the same with $D = \varepsilon I + A$.
- $\|Res\|_* \le \min\{\eta_{dir}, \eta_A + \eta_S\}$, where $\eta_S \ge \|S^{\sim}\|_*$ comes from gaps 2 and 6.
- Then:
  - $|||e||| \le \min\{\dots\} + \eta_{bdry}$;
  - $|||e|||_{\oplus} \le 3(\min\{\dots\} + \eta_N + \eta_D) + |||w||| + \eta_{DA}$;
  - $E \le$ the same $+ \eta_S$.
- The boundary terms are those of gap 4 (non-P1 Dirichlet data, Neumann oscillation).

4. What is proved, and how

(i) Guaranteed upper bound: complete proof, elementary.

- For $v \in H^1_D$ and any $\sigma_h \in H(\mathrm{div})$ with $\sigma_h \cdot n = -\Pi_F g_N$ on Γ_N:

$$Res(v) = (r - \Pi_1 r, v) + (\Pi_1 r - \mathrm{div}\,\sigma_h, v) - (\tau, \nabla v) + (g_N - \Pi_F g_N, v)_{\Gamma_N}.$$

- Each term is bounded cell by cell:
  - $(\tau, \nabla v)_K$ uses either Cauchy–Schwarz, or the identity $(\tau, \nabla v)_K = (\tau \cdot n, v - \bar{v}_K)_{\partial K} - (\mathrm{div}\,\tau, v - \bar{v}_K)_K$ together with the trace inequality of gap 4 and Poincaré;
  - the second term uses $\|v\| \le \sigma^{-1/2}|||v|||$;
  - the first uses Poincaré with $\min\{h/(\pi\sqrt{\varepsilon}), \sigma^{-1/2}\}$.
- No compatibility, no Galerkin orthogonality and no algebraic-error term are needed. So the bound holds for any u_h, including inexact AFC iterates. This is SV20 Thm. 3.1, with their weighted flux term $w_K \varepsilon^{-1/2}\|\tau\|$ replaced by the computable min above; up to inverse-inequality constants, the min is $\le$ their term.

(ii) Efficiency: proof sketch, not refereed.

- **Patch stability.** For the A-version at the discrete solution, $Res_A(\psi_a) = 0$, because the stabilisation is exactly $(A\nabla u_h, \nabla v_h)$. So SV20 Lemma 4.2 goes through with $\varepsilon^2 \nabla u_h \to D\nabla u_h$.
  - Their two comparison fluxes need only P_1 data and $D\nabla u_h \in P_1^d \subset RTN_1$.
  - With $w_a = 1$, the minimiser satisfies $\|\tau^a\|^2 \lesssim h\|j\|^2 + (\varepsilon/\sigma)\|r\|^2$. The trace-type evaluation then gives $(\sigma\varepsilon)^{-1/4}\|j\| + \sigma^{-1/2}\|r\|(\sqrt{(\varepsilon/\sigma)}/h)^{1/2}$, which is below the robust residual terms in the reaction-dominated regime.
  - So the weights are needed in the estimate (SV20 Prop. 5.1), not in the construction.
- **Lower bound for the residual estimator.** The residual estimator of Res_A is $\lesssim \|Res_A\|_* +$ oscillation. This is the bubble argument of Verfürth (SINUM 43, 2005) Thm. 4.1, which tests only the residual functional and so holds for any u_h. His lower bound is global: the dual norm of the convective derivative does not localise.
- **Chain.** $\|Res_A\|_* \le \|Res\|_* + \|S^{\sim}\|_*$, and $\|Res\|_* \lesssim |||e||| + \|b\cdot\nabla e\|_*$ (Verfürth Lemma 3.1, under his assumption $\|c\|_\infty \le c_b \sigma$).
- **Result.** $\eta_A \lesssim |||e|||_{\oplus} + \|S^{\sim}\|_*$ + oscillation, globally, with constants independent of ε, σ, $|b|$ and h.
- **Open:**
  - the efficiency of η_S ($\|S^{\sim}\|_*$ bounds, gap 2);
  - local (rather than global) efficiency;
  - $\sigma = 0$ (there `esv/robust` are used, robust numerically in gap 2).

5. Differences from SV20

SV20	galerkit
$-\varepsilon^2\Delta u + \kappa^2 u = f$, Galerkin u_h	$-\varepsilon\Delta u + b\cdot\nabla u + cu = f$, any conforming u_h (all algebraic stabilisations, inexact iterates); the convection enters only through the data $f^{\sim}$
weights w_a in the construction	$w_a = 1$ by default (sharper in tests: see §6); SV's weights are available via `w_const`
flux term $w_K \varepsilon^{-1/2}\|\tau\|$ with C^* from inverse inequalities	computable $\min\{\varepsilon^{-1/2}\|\tau\|, \Sigma\, c_F\|\tau\cdot n\| + m_K\|\mathrm{div}\,\tau\|\}$, with no inverse-inequality constants

SV20	galerkit
Dirichlet u = 0	non-P1 Dirichlet data (gap 4 lifting), Neumann data (prescribed normal flux + oscillation)
any p	p = 1 (P1 elements, RTN_1 patches)
energy norm	energy norm, augmented norm, and E (A-version + η_S)

6. Numerics (smoke runs; the full studies are yours to run)

Uniform meshes n = 8, 16 and 32; BJK and SMUAS; σ = c. The reference norms use P2 Riesz representatives on the same mesh, which are lower bounds, so effectivities are slightly overestimated.

problem	ε	augmented 3η/|||e|||_⊕: SV (this)	gap 6 hybrid	classic ESV
smooth, c = 1	10^{-2}	3.4–4.0	4.1–5.0	3.9–4.8
smooth, c = 1	10^{-4}	5.4–5.9	4.6–8.2	6.2–28
smooth, c = 1	10^{-6}	4.6–7.3	4.3–6.5	37–93
smooth, c = 1	10^{-8}	4.5–6.6	4.3–6.4	112–278
outflow layers, c = 1	10^{-2} … 10^{-8}	3.2–4.4	3.2–4.1	3.3–223
smooth, c = 10^{-3} … 10^{-1}, n = 16	10^{-4}, 10^{-6}	4.4–5.0	3.8–5.2	4.2–24
3D outflow layers, c = 1	10^{-2}, 10^{-6}	3.3–4.0	3.3–4.5	4.5–132

- **Bound check:** the upper bounds held in every case, including an iterate after 5 nonlinear steps, Neumann data and 3D.
- **E effectivity:** 2.9–13. The top of that range is an artifact of the same-mesh reference for ‖S~‖_* (gap 6, nested-mesh check).
- **Weight in the construction:** w_a = 1 gives 3.2–5.5; SV's weight with C = 1 gives 4.3–14 (scan over C = 0.3, 1, 3, 10, 30, ∞ on the same cases: the larger C, the sharper).
- **Cost:** 0.3 s for a 2D mesh with 2048 cells; 5 s for 3072 tetrahedra (sparse LU of all patches together).

7. Status against the literature, and what is publishable

- **SV20:** reaction–diffusion only.
- **Verfürth 2005; Tobiska & Verfürth (arXiv 1402.5892):** robust residual estimators for convection–diffusion and for stabilised methods, in the same norm. Their constants are unknown, and the upper bound contains a consistency error of the stabilisation.
- **ESV10:** guaranteed, for DG; robust only up to cut-off ratios.
- **Eigel & Merdon 2016:** guaranteed, in the augmented norm. SV20 Prop. 5.1 shows that it cannot be robust for reaction dominance.
- **Cai & Cai 2021:** robust hybrid estimators with unknown constants.
- **What this gives:**
  - a guaranteed upper bound with no unknown constants and no consistency term, valid for every algebraic stabilisation and at inexact iterates;
  - robust numerically in the augmented norm;
  - with a proof sketch of global robust efficiency in E.
- **Literature check:** I found no published result with all of these. That is based on quick searches only; run `/litscan` before claiming novelty.
- **Before submitting, three things are needed:**
  1. a full proof of the efficiency sketch in §4(ii), and ideally an efficiency proof for η_S;
  2. the full studies (aj27/abr17/guaranteed3d with `--estimator sv`);
  3. the literature scan.

8. Correction to gap 5 (from the uploaded Knobloch 2017)

- The Kuzmin DMP condition $\min\{a_{ij}, a_{ji}\} \le 0$ is Knobloch's Theorem 2 (14).
- galerkit's `dmp_fix` for Kuzmin (both-node limiting) is his Limiter 2.
- The gap 5 docs and `dmp.py` now credit him.
- What remains galerkit's own in gap 5:
  - the 3D newest-vertex-bisection analysis (the obtuse level $\equiv 1$ class);
  - the F-BBK condition and weight floor;
  - the numerics.

9. Update: $\sigma \ge 0$ and an equilibrated bound for $S^{\sim}$ (gaps 2 and 3 of the October 2026 scan)

What changed in `sv_estimator`:

- **Defect correction.** Some patches have no free (Dirichlet) face. On each of them the patch data g_a is shifted by $\delta_a/|\omega_a|$, where $\delta_a = R(\psi_a)$, which makes the local problems compatible.
  - For Res_A, δ_a is minus the algebraic residual, so it vanishes at the discrete solution.
  - For $S^{\sim}$, $\delta_a = -S(u_h; u_h, \psi_a)$.
- **$\sigma = 0$.** The patch problems are solved with the exact constraint $\operatorname{div} v = g_a$: Braess–Schöberl with RTN_1, with one redundant row dropped per compatible patch. For $\sigma > 0$ the relaxed SV problem is kept.
- **Spread defect.** The defect $\psi_{def} = \Sigma_a \delta_a \chi_{\omega_a}/|\omega_a|$ is piecewise constant. It is bounded by one global RT_0 mixed solve (`_defect_bound`): $\min\{\|\varepsilon^{-1/2}s\|, \sigma^{-1/2}\|\psi_{def}\|\}$, with $\operatorname{div} s = \psi_{def}$ and $s\cdot n = 0$ on Γ_N. This needs no Friedrichs constant, and the Neumann/$\sigma = 0$ caveat of the old η_{alg} no longer applies. At inexact iterates it is the algebraic-error term.
- **"S" mode.** $-S^{\sim}(v) = -(A\nabla u_h, \nabla v)$ is equilibrated by the same patch machinery, with $r = 0$, $D = A$ and $g_N = 0$. The result is $\eta_S = \min\{\text{gap 2/6 bound, equilibrated bound}\}$.
- **Bound.** $\|\mathrm{Res}\|_* \le \min\{\eta_{dir}\ (\sigma > 0 \text{ only}), \eta_A + \eta_S\}$. The bounds on $|||e|||$, $|||e|||_{\oplus}$ and E are as in §3. `estimator="sv"` now accepts $\sigma = 0$.

Why the new η_S is efficient (sketch).

- **Defect part.** It is the functional $v \mapsto S^{\sim}(Iv)$ with $Iv = \Sigma_a$ (mean over ω_a of $v)\psi_a$. Since $|||Iv||| \le C_I|||v|||$, its dual norm is $\le C_I\|S^{\sim}\|_*$. The RT_0 min-norm flux computes the dual norm of this P0 functional up to the stability constant of the mixed method (for $\sigma = 0$; for $\sigma > 0$ the min with $\sigma^{-1/2}\|\psi\|$).
- **Patch part.** After the shift the data are compatible, and SV20 Lemma 4.2 bounds the patch part by the jumps of $A\nabla u_h$. The constant shift drops out for mean-zero test functions. By Verfürth's face bubbles, $S^{\sim}(b_F w) = ([[A\nabla u_h\cdot n]], b_F w)_F$, so $\varepsilon^{-1/2}\alpha_F^{1/2}\|[[A\nabla u_h\cdot n]]\|_F \lesssim \|S^{\sim}\|_{*,\omega_F}$.
- **Status.** Together with §4(ii), this gives a sketched two-sided bound in E for both the A-version and η_S. It is not refereed.

Numerics (rectangle_mesh(16); BJK and SMUAS; references on the same mesh):

	ε	augmented $3\eta/\|\|\|e\|\|\|_{\oplus}$: SV	ESV	E: SV	gap 2 `robust`	$\eta_S/\|S^{\sim}\|_*$: equilibrated	gap 2
smooth, c = 0	10^{-2} … 10^{-8}	3.8–4.7	3.8–4.5	3.0–3.8	3.5–4.5	1.18–1.30	1.20–1.44
outflow layers, c = 0	10^{-2}, 10^{-6}	3.6–4.8	3.2–3.8	2.9–3.3	3.5–3.6	1.25–1.26	1.22–1.44
smooth, c = 0, 5 nonlinear steps	10^{-6}	5.3	8.7	4.1	7.5	1.30	1.25
smooth, c = 1	10^{-2} … 10^{-8}	3.7–6.2	4.0–6.1	3.2–10.2	3.4–10.1	1.5–17	1.4–15

- **$\sigma = 0$.** The SV estimator is guaranteed and robust, and sharper than the gap 2 `robust` estimator in E. The equilibrated η_S is within 30 % of $\|S^{\sim}\|_*$.
- **$\sigma > 0$, small ε.** Both $\eta_S/\|S^{\sim}\|_*$ ratios grow. This is the known same-mesh reference artifact (gap 6, nested-mesh check), and the smaller of the two bounds is used.

- **Tests.** `tests/test_cdr_sv_estimator.py`, in particular `test_sigma_zero` and `test_equilibrated_stabilisation_bound`.

10. Update: the efficiency proof (gap 1 of the October 2026 scan)

The full proof is in sv_efficiency_proof.md (PDF alongside). It supersedes the sketch in §4(ii) and §9.

- **Correction 1.** The §4(ii) sketch bounded only the flux term. With w = 1, the equilibration term needs a different competitor: the best-approximation trick of Lemma G.
- **Correction 2.** The augmented-norm factor is κ = 2 + max{1, c_b}, not 3, unless c − ½ div b ≡ σ. The code now uses κ (`kappa_aug`), which changes the augmented bounds of AJ27 Ex. 4.3.
- **Correction 3.** The σ > 0 defect term is now the joint minimum min_s (ε^{-1}‖s‖² + σ^{-1}‖ψ − div s‖²)^{1/2}. This is never larger than the earlier min{·, ·} form, and it is provably quasi-optimal.

11. Update: local efficiency in a localisable measure (gap 4 of the October 2026 scan)

The theory is in sv_efficiency_proof.md §9.

- **The measure:** E_loc² = |||e|||² + Σ_a(‖B_A'(e, ·)‖²_{,a} + ‖S~‖²_{,a}), with local dual norms on the vertex patches.
- **The bounds:**
  - E_loc ≤ √(d+1)E ≤ `eta_Eloc` (guaranteed);
  - local efficiency η_K ≲ E_loc(ω~_K) + osc;
  - E ≲ E_loc + η_{def,S}.
- **Code:**
  - `flux_estimator.localized_error(S, u, σ, A∇u_h)` computes the reference E_loc (patch P2 Riesz representatives);
  - `AdaptiveAFC(..., estimator="sv", loc_error=True)` adds the rows `err_loc`, `eta_sv_loc`, `eff_loc` and `eta_def_S`.

12. Update: the 3D study of the SV estimator (item 5 of the October 2026 scan)

What was implemented. Two new studies in `galerkit.convection_diffusion.guaranteed3d`, run by `examples/convection_diffusion/3d/guaranteed_estimators/run_sv.py`:

- `sv_sweep`**:** robustness in ε. It uses the problems, ε values and uniform Kuhn meshes of the existing `sweep`:
  - problems `bubble0` (σ = 0), `bubble1` (σ = 1) and `layers0` (σ = 0);
  - ε = 10^{-1} ... 10^{-8} and n = 2 ... 16;
  - SV marks and reports, with ESV and the gap 2 bound recorded alongside;
  - the localised measure E_loc is on by default.
- `sv_adaptive`**:** SV-marked adaptive runs (newest-vertex bisection, dual marking) on `layers` (σ = 1), `layers0` (σ = 0) and `abr66` (σ = 0). E_loc is computed only with `--loc-error`.
- **Jobs:** 144 (eight methods), picked up automatically by `scripts/server/run_guaranteed3d.sh`.
- **New rows:** `eta_sv_S`, `eta_sv_S_eq`, `eff_S_sv` (η_S/‖S~‖_*) and `kappa_aug`.
- **New tables:** `tables.md` reports min–max effectivities per ε and method for the augmented norm, E, S~ and E_loc, together with κ and η_def,S/E_loc.
- **Test:** `test_sv_studies` in `tests/test_cdr_guaranteed3d.py`.

Smoke numbers only (BJK and SMUAS, n ≤ 4 and ≤ 10^3 dofs; these are not results). All bounds held in every loop.

problem	ε	κ SV/|||e|||_⊕	SV_E/E	η_S/‖S~‖_*	SV_Eloc/E_loc	η_def,S/E_loc
bubble0 (σ = 0)	10^{-2}, 10^{-6}	5.4–6.8	4.1–6.3	1.25–1.44	10–13	≤ 0.42
layers0 (σ = 0)	10^{-2}, 10^{-6}	4.6–5.2	3.3–3.6	1.23–1.29	8.6–10	≤ 0.31

problem	ε	κ SV/|||e|||_⊕	SV_E/E	η_S/‖S˜‖_*	SV_Eloc/E_loc	η_def,S/E_loc
bubble1 (σ = 1)	10^{-2}	4.1–5.3	3.1–4.8	1.7–1.9	8.1–10	≤ 0.28
bubble1 (σ = 1)	10^{-6}	3.8–4.5	6.4–9.6	13–18	11–18	≤ 0.14
adaptive layers, layers0, abr66	10^{-2}, 10^{-4}, 10^{-3}	4.0–5.7	3.0–3.8	1.15–1.69	—	—

What the smoke runs show.

- The 3D picture matches the 2D one. For σ = 0 the SV estimator is guaranteed and ε-independent, and slightly sharper in E than the gap 2 bound.
- SV_Eloc/E_loc is about twice SV_E/E, as the √(d+1) = 2 factor predicts.
- **σ > 0, small ε.** η_S/‖S˜‖_* reaches 13–18. This is most likely the same-mesh P2 reference artifact of §11: the dual norm ‖S˜‖_* is under-resolved when √(ε/σ) ≪ h. The nested-reference check was 2D only, so the full runs should repeat it in 3D before this growth is attributed to the estimator.

13. Update: adaptive stopping in E (item 6 of the October 2026 scan)

What was implemented.

- `sv_estimator.sv_algebraic(S, op, u, b, σ)` returns η_{def,A} from the algebraic residual alone. It needs one global RT_0 solve and no patch problems. It agrees with the full estimator's η_{def,A} to quadrature accuracy.
- `EstimatorStop(kind="sv")` stops when κη_{def,A} ≤ γ η_disc, with η_disc = κ(η_A − η_{def,A}) + η_S + bd:
  - every check computes only `sv_algebraic`;
  - the full SV estimator is evaluated only for candidates, on the existing growth schedule.
- `AdaptiveAFC(estimator="sv", adaptive_stop=γ)` uses this rule by default. Rows gain `stop_eta_E`.
- **In the 3D studies**, `guaranteed3d run --adaptive-stop 0.1` for the `sv_*` studies picks it up automatically.
- **Theory:** sv_efficiency_proof.md §10, Proposition 5:
  - E(u^i) ≤ η_disc + η_alg at every iterate;
  - E ≤ (1 + γ)η_disc at the stopped iterate;
  - for γ small, the bound is two-sided and robust.

Smoke result (2D, ε = 10^{-6}, σ = 0, BJK). With γ = 0.1 the iteration stops after 10–15 iterations instead of hitting 10 000 without converging. E is smaller at the stopped iterate, and η_E/E = 4.0–4.5.

Timing log (the earlier request). Every row now records `solve_time`, `est_time`, `err_time`, `aug_time` and `loc_time`, and the loop log prints them.

- Measured at 729 vertices in 3D: E_loc costs 2.1 s out of a 16.6 s loop (solve 6.1 s, estimators 6.4 s, all reference errors 4.1 s).
- The earlier "15 s for E_loc" was the whole loop. E_loc grows linearly in the number of vertices, so about 15 s at n = 16.

14. Update: variable diffusion ε(x) (the remaining gap 7 limitation)

What was implemented.

- `sv_estimator`**:**
  - the fluxes use D = $\Pi_1\varepsilon$ I + A, with $\Pi_1\varepsilon$ the cellwise L^2 projection of ε onto P1, so D∇u_h is a P1 field per cell;
  - the patch problems, the residual and the τ norms take that P1 field;
  - η_ε,K = ‖ε^{-1/2}(ε − $\Pi_1\varepsilon$)∇u_h‖_K is added to the indicators and returned as `eta_eps`;
  - ε^{-1/2}‖τ‖ is evaluated with ε pointwise, and the cell minimum ε_lo,K enters every inverse and trace constant and the defect solve;
  - `sv_algebraic` is unchanged in substance, since P1 tests do not see ε − $\Pi_1\varepsilon$.
- `abr17_estimator` (ABR17, ESV, gap 2 `robust`, which the SV estimator also reads):
  - it now adds the cell-mean remainder η_ε,K = ‖ε^{-1/2}(ε − $\bar{\varepsilon}$_K)∇u_h‖_K;

  - it uses ε_lo,K in the ε^{-1/2} weights, and the cell maximum ε_hi,K in the ε^{1/2} weights of the Dirichlet-lifting terms;
  - before this change it only warned "eps varies: cell means used", and the bound was not guaranteed.
- **Reference errors.** `energy_error`, `augmented_error` and `localized_error` now use ε(x) at the quadrature points. Before, the energy norm used max ε.
- **New problem:** `get_problem("variable_eps", dim, eps, contrast, kind="smooth" | "jump", b, c)`:
  - `smooth`: ε = ε₀ k^{x₁};
  - `jump`: ε₀ and ε₀k on either side of x₁ = ½, with continuous flux;
  - manufactured exact solution in 2D and 3D.
- **Constant ε:** results are unchanged, to rounding.
- **Theory:** sv_efficiency_proof.md §11, Theorem 6. Reliability needs only ε ≥ ε₀ > 0. Local efficiency has constants that depend on the patch contrast.

Smoke results.

- All bounds held in 48 2D cases, four 3D cases and two adaptive loops.
- For smooth ε at contrast 10^4, κ SV/|||e|||_⊕ = 3.5–4.3, against 8.3–8.9 with cell means.
- For an aligned jump, the effectivity does not depend on the contrast up to 10^4.
- Robustness in jumps is not proved, and non-quasi-monotone (checkerboard) ε is untested.